Lokesh Reddy Sodanapalli

Embedded System Test & Validation Engineer

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PROFESSIONAL EXPERIENCE

01/2026 – Present
Dresden, Germany

Test Automation Engineer, System Test - Working Student, *Joynext GmbH*
- Developed Python-based automation libraries and Robot Framework keywords for embedded automotive ECU testing (SSH, UART).
- Designed and executed automated system validation workflows for RTCU lifecycle and power management (including low-power modes).
- Took ownership of automation development for embedded automotive testing, including Python libraries, Robot Framework test cases, and test planning activities
- Worked with automotive communication protocols: Automotive Ethernet, CAN, UDS, DoIP, TCP/IP, Serial.
- Conducted log analysis and debugging (Linux) using DLT tools and system logs.
- Contributed to requirement analysis and transformation into structured test cases and validation strategies.

12/2024 – 07/2025
Berlin, Germany

Software Test - Working Student, *Mbition GmbH*
- Performed system, integration, and regression testing for Mercedes-Benz infotainment domain controller.
- Developed automated test scripts using Python and Robot Framework.
- Conducted network-level testing using Automotive Ethernet and TCP/IP.
- Performed bug analysis, validation, and log-based debugging.
- Worked in Linux and QNX environments with CI/CD pipelines.
- Managed test documentation and defect tracking using Jira and Confluence.

05/2024 – 10/2024
Ulm, Germany

System Integration and Test Engineering Intern, *Continental ADC GmbH*
- Developed and integrated custom HIL-like test bench setups for ADAS radar sensors and ECUs.
- Performed system-level validation of radar-based ADAS features under controlled test conditions.
- Conducted online and offline calibration of sensor positioning and alignment(ADAS Camera).
- Automated flashing, logging, and test execution using Python (SSH-based workflows).
- Debugged system-level performance issues and reduced ECU CPU load from 90% to ~25% through root-cause analysis.
- Worked with automotive communication protocols: CAN, Automotive Ethernet.
- Supported system verification and validation activities across hardware and software integration layers.

06/2021 – 08/2022
Bengaluru, India

Assistant System Engineer, *Tata Consultancy Services Ltd.*
- Performed infotainment system validation for Jaguar Land Rover (JLR) in an Agile development environment.
- Worked with hardware test rigs for infotainment system testing and validation.
- Conducted functional, acoustic, and non-functional testing, including speech and Alexa-integrated features.
- Planned and executed structured test cycles and defect tracking using Jira.
- Collaborated closely with JLR engineering teams to ensure system quality and correct functionality.

CERTIFICATIONS

06/2024 – 08/2024
Ulm, Germany

A AN: CANoe Basic, *Continental Learning Platform*
- Ich habe mir fundierte Kenntnisse in den Grundlagen von Vector CANoe angeeignet, einschließlich Messaufbau, Simulation, Diagnoseanalyse und CAPL.

SKILLS

Programming / Scripting Languages ●●●○○
Python (Object-Oriented Programming) C, C++, Linux, **Matlab/Simulink**, Git, MS Office Suite, Robot Framework, Machine Learning

Soft Skills ●●●●○
Analytical Thinking, Teamwork, Communication Skills, Commitment, Professionalism, Support, Innovation, Solution Orientated, Quality Awareness

Automotive Technology ●●●●○
Automotive Tools: ADAS Concepts, CANoe, CANdb++, ODIS, ET Framework, GitLab CI, Jenkins, Jira, Confluence, Polarian
Communication Protocols: Automotive Ethernet, CAN, CAN-FD, FlexRay, TCP/IP, Serial (UART), P-CAN

Hardware ●●●○○
Measurement & RF Tools: Oscilloscope, Spectrum Analyzer, Signal Generator, Data Evaluation, Test Bench Design
Electrical Engineering: Digital Signal Processing, Embedded Systems, Communications Engineering, Control Systems, Circuit Design

Testing & Validation
Test Case Creation, Error Analysis, Test Automation & Scripting, Regression Testing, ADAS Testing, Test planning, Verification testing, Test documentation and reporting, System validation

EDUCATION

10/2022 – 05/2026
Chemnitz

Master's in Embedded Systems, *Technische Universität Chemnitz*
Current Grade: 2.5
Focus: Hardware Software-Codesign | Digital Systems | Automotive communications | Real Time Operating Systems | Sensor Systems | Digital Signal Processing

Master's Thesis: Validation of Ego-Motion Algorithm using Synthetic Radar Data
- Developed validation framework in Python
- Evaluated algorithm performance (velocity, yaw-rate, robustness)
- Worked with synthetic radar data (MATLAB Automotive Toolbox)

07/2017 – 06/2021
Chennai, India

Bachelor's in Electrical and Electronics Engineering, *Vellore Institute of Technology*
Final Grade: 2.3
Focus: Microprocessor | PowerSystems | Semiconductor technology | Control Theory | Digital and Analog devices | Object Oriented Programming (OOPS)

Bachelor's Thesis: Design of a Home Energy Management System to Maximise Battery Utilisation for Residential Installation.
- Developed a MATLAB Simulink model for a grid-connected PV system with integrated battery management.
- Optimised battery utilisation and improved overall energy efficiency by implementing control algorithms and validating simulation outcomes.

UNIVERSITY PROJECTS

04/2023 – 09/2023

Study and implementation of AC analysis methods in STM32, *Master's Project*
- Developed DSP algorithms (Goertzel, DFT, DTFT) on STM32 microcontrollers and analysed time-series data for performance evaluation.
- Implemented DFT and Goertzel algorithms for real-time analysis and validation on microcontroller hardware.

07/2020 – 10/2020

Intelligent Lighting System, *(LDR Sensor, PCB Design & Soldering), Bachelor's Project*
- Contributed in circuit design and participation in project implementation,focusing on the functionality of the LDR sensor and PCB development.

11/2019 – 05/2020

Line Following and Obstacle Avoidance Robot, *Bachelor's Project*
• Participated in the hardware development for a line-following robot usingArduino, with primary responsibility for hardware design and implementation.

LANGUAGES

English ●●●●○
IELTS C1

Deutsch ●●●○○
DaF B1 (in progress toward B2)